

Caching Django websites with Memcached

Django is an open source Web application framework written in Python. When used in conjunction with Memcached, it dramatically improves the performance of heavily used websites and reduces database access. Read on to learn how to set up and configure Memcached for a Django website.



Does your Django website's performance slow down due to heavy loads or a series of requests and responses between the client and the Web server that hosts the requested site? Or do you wish you could handle your medium or high-traffic Django websites better?

In which case, what you probably need is a caching system, or a Web caching system.

Web caching

Web caching means storing a copy of Web objects, new to the user, that would otherwise be slower to access/generate. Examples of some Web objects are Web pages (the HTML itself), images in Web pages, etc. Doing this will reduce bandwidth needs and costs, while also reducing the load on the website server.

Memcached

Memcached is a memory-based caching (mem - cached) system with a specific amount of RAM allocated to it. This

is the most efficient type of cache available to Django. It runs as a daemon and can share cache over multiple servers. Therefore, even when running the daemon over multiple machines, it still acts as a single cache.

Memcached installation

1. `apt-get install memcached`
2. `apt-get install python-memcache`

The first command installs Memcached and the second one installs the Python API for communication between the application and the Memcached daemon. Memcached runs on Port 11211 on localhost (127.0.0.1) with default configuration. This configuration can be modified in the file `/etc/memcached.conf`

Configuring Django

The cached data stays in the memory. The `setting.py` file has to be modified.